INSTITUTE OF TECHNOLOGY OF CAMBODIA
DEPARTMENT OF APPLIED MATHEMATICS AND STATISTICS

Final Project Report:
Garman-Kohlhagen European FX Option Pricing

Lecturer:Dr. LUEY Sokea (Course & TP) **Class: Partial Differential Equation**

GROUP: 3

Name of Student	ID of Student	Score
HEM Bellyday	e20220800	…….
LENG Devid	e20220888	…….
PHAL Menghak	e20220544	…….
LOT Soklang	e20221464	…….

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Abstract

This project presents a comprehensive study of European currency option pricing under the Garman–Kohlhagen extension of the Black–Scholes framework. The exchange rate dynamics are modeled using a stochastic differential equation, from which the corresponding partial differential equation (PDE) is derived through Itô calculus and no-arbitrage arguments. The PDE is solved analytically by transformation to the classical heat equation, yielding closed-form pricing formulas that serve as benchmark solutions.

To complement the analytical results, numerical solutions are obtained using finite difference methods, including the Forward-Time Central-Space (FTCS), Backward-Time Central-Space (BTCS), and Crank–Nicolson schemes. The stability, convergence, and accuracy of these schemes are examined through theoretical analysis and computational experiments. Numerical prices and Greeks are validated against analytical values, demonstrating high accuracy and confirming the reliability of the numerical framework.

The study further investigates practical implications through sensitivity analysis and delta-hedging simulations, showing significant risk reduction when Greeks are used for hedging. Overall, the results establish a rigorous connection between stochastic modeling, analytical solutions, and numerical approximation, providing a solid foundation for PDE-based currency option valuation and risk management.

Contents

1 Introduction

This project investigates the valuation of European currency options within the Black–Scholes framework, focusing on the Garman–Kohlhagen extension that explicitly incorporates both domestic and foreign interest rates. The model adapts the classical Black–Scholes methodology to foreign exchange markets by treating the foreign interest rate as a continuous yield, reflecting the return earned on holding foreign currency.

The study develops the theoretical foundation of the model, derives the governing partial differential equation from stochastic exchange rate dynamics, and obtains analytical pricing formulas. In addition, numerical finite difference methods are implemented to approximate the solution, and their stability, convergence, and accuracy are evaluated against the analytical benchmark. This combined analytical and computational approach provides a comprehensive understanding of currency option pricing and its practical implications for risk management.

2 Literature Review

The foundational work on currency option pricing was established by Garman and Kohlhagen (1983), who extended the Black–Scholes option pricing model to account for two distinct interest rates — domestic and foreign — inherent in foreign exchange markets. Their closed-form formulas for European call and put currency options remain a benchmark in both academic research and practical applications. :contentReference[oaicite:0]index=0

Subsequent research has explored the limitations and extensions of the Garman–Kohlhagen framework. For example, Prabakaran et al. (2026) compare the Garman–Kohlhagen model with the Heston stochastic volatility model and find that models incorporating stochastic volatility often fit real FX option data more accurately than the constant-volatility Garman–Kohlhagen approach. :contentReference[oaicite:1]index=1

Other studies have investigated alternative dynamics and market features that the classical model does not capture. Krylova (2022) discusses volatility term structures implied by FX option prices and highlights how market conventions based on the Garman–Kohlhagen model are used to construct arbitrage-free volatility surfaces. :contentReference[oaicite:2]index=2

Additional work has examined revisions to the basic model to incorporate uncovered interest parity and mean-reversion, indicating ongoing efforts to generalize and refine currency option pricing theory beyond the original Garman–Kohlhagen assumptions. :contentReference[oaicite:3]index=3

Together, this literature establishes the relevance of the Garman–Kohlhagen model as both a theoretical foundation and a benchmark for developing more advanced pricing frameworks in foreign exchange option markets.

3 Objectives

The main objective of this project is to value European call and put currency options using the Black–Scholes (Garman–Kohlhagen) model. To support this goal, several supporting objectives are included. These involve understanding why two interest rates (domestic and foreign) are required in currency option pricing, explaining the economic meaning of the model parameters, and formulating the exchange rate dynamics using a stochastic differential equation (SDE). From this SDE, the corresponding Black–Scholes partial differential equation (PDE) is derived and solved to obtain the analytical pricing formulas for European currency options. In addition to the theoretical derivation, the project also applies numerical methods, including simulation techniques such as Monte Carlo, to compute option prices and compare numerical results with the analytical solution in order to evaluate accuracy and pricing errors.

4 Research Questions

This project is guided by the following research questions:

- How can European currency options be valued using the Black–Scholes (Garman–Kohlhagen) framework?
- Why are two interest rates (domestic and foreign) necessary in currency option pricing, and what is their economic interpretation?
- How does the stochastic differential equation (SDE) describing exchange rate dynamics lead to the Black–Scholes partial differential equation (PDE)?
- How is the PDE solved to obtain the analytical pricing formulas for European call and put currency options?
- How do numerical simulation methods, such as Monte Carlo simulation, compare with the analytical solution in terms of pricing accuracy and error?
- What are the main sources of numerical error when approximating option prices using simulation methods?

5 Conceptual Background of Currency Options

Before developing the mathematical model, it is important to understand the financial and economic concepts behind currency options and foreign exchange markets. This section introduces the theoretical foundation needed for pricing currency derivatives.

5.1 Foreign Exchange Market

The foreign exchange (FX) market is where currencies are traded. The exchange rate represents the price of one currency in terms of another. Exchange rates fluctuate due to supply and demand, interest rates, inflation, economic performance, and political stability. Because exchange rates are uncertain and change continuously, they are often modeled as stochastic processes in financial mathematics.

5.2 Currency Options

A currency option is a financial derivative that gives the holder the right, but not the obligation, to buy or sell a currency at a predetermined exchange rate (strike price) on or before a specified maturity date. A call option gives the right to buy the foreign currency, while a put option gives the right to sell it. These instruments are widely used for hedging exchange rate risk and for speculation.

5.3 Role of Two Interest Rates

Unlike equity options, currency options involve two risk-free interest rates: the domestic interest rate and the foreign interest rate. The domestic rate reflects the return on investments in the home currency, while the foreign rate represents the return on the foreign currency. In currency pricing theory, the foreign interest rate plays a role similar to a dividend yield in stock option models. The difference between the two rates influences the forward exchange rate and therefore affects option valuation.

5.4 Exchange Rate Risk and Hedging

Firms and investors face exchange rate risk when cash flows are denominated in foreign currencies. Currency options provide a flexible hedging tool, allowing protection against unfavorable exchange rate movements while preserving the opportunity to benefit from favorable changes.

5.5 Need for Mathematical Modeling

Because exchange rates evolve randomly over time, deterministic models are not sufficient. Stochastic modeling allows the exchange rate dynamics to be described using probability theory. This leads to the formulation of a stochastic differential equation (SDE), which forms the foundation for deriving the pricing partial differential equation (PDE) used in the Black–Scholes framework.

6 Model Formulation and Analytical Solution

6.1 From Stochastic Differential Equation to Pricing PDE

Let S_t denote the domestic price of one unit of foreign currency. Under the risk–neutral measure, the exchange rate follows a geometric Brownian motion

$$dS_t = (r_d - r_f)S_t\,dt + \sigma S_t\,dW_t,$$

where r_d and r_f are the domestic and foreign risk–free interest rates, σ is the volatility, and W_t is a standard Brownian motion.

Let the option value be $V(S,t)$. By Itô's Lemma,

$$dV = V_t\,dt + V_S\,dS + \frac{1}{2}V_{SS}(dS)^2.$$

Since $(dS)^2 = \sigma^2 S^2 dt$, substitution gives

$$dV = \left(V_t + (r_d - r_f)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \sigma S V_S dW_t.$$

Consider the portfolio $\Pi = V - V_S S$. Its differential is

$$d\Pi = dV - V_S dS,$$

which yields

$$d\Pi = \left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)dt.$$

The stochastic term is eliminated, so the portfolio is risk–free. By no–arbitrage,

$$d\Pi = r_d \Pi\,dt = r_d(V - SV_S)dt.$$

Therefore,

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} = r_d V - r_d S V_S,$$

and the Garman–Kohlhagen PDE follows:

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r_d - r_f) S V_S - r_d V = 0.$$

6.2 Analytical Solution of the Pricing PDE

Let $\tau = T - t$ denote time to maturity. Then $V_t = -V_\tau$, and the PDE becomes

$$V_\tau = \frac{1}{2}\sigma^2 S^2 V_{SS} + (r_d - r_f) S V_S - r_d V.$$

Introduce the transformation

$$V(S,t) = e^{-r_d \tau} F(S,\tau),$$

which leads to

$$F_\tau = \frac{1}{2}\sigma^2 S^2 F_{SS} + (r_d - r_f) S F_S.$$

Using the logarithmic change of variable $x = \ln S$, the derivatives transform as

$$F_S = \frac{1}{S} F_x, \quad F_{SS} = \frac{1}{S^2}(F_{xx} - F_x),$$

and the PDE becomes

$$F_\tau = \frac{1}{2}\sigma^2 F_{xx} + \left(r_d - r_f - \frac{1}{2}\sigma^2\right) F_x.$$

Let

$$F(x,\tau) = e^{\alpha x + \beta \tau} u(x,\tau),$$

with $\alpha = -\dfrac{r_d - r_f - \frac{1}{2}\sigma^2}{\sigma^2}$, reducing the equation to the heat equation

$$u_\tau = \frac{1}{2}\sigma^2 u_{xx}.$$

Solving under the terminal payoff condition $V(S,T) = \max(S - K, 0)$ yields the Garman–Kohlhagen formula for a European call:

$$C = S_0 e^{-r_f T} N(d_1) - K e^{-r_d T} N(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r_d - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

7 Methodology

This study adopts both analytical and numerical approaches to value European currency options under the Garman–Kohlhagen framework. The methodology is designed to establish a theoretical benchmark and then evaluate numerical approximations against it.

7.1 Analytical Benchmark Solution

The first step consists of solving the Garman–Kohlhagen partial differential equation (PDE) analytically. By applying a sequence of variable transformations, the PDE is reduced to the classical heat equation. The known solution of the heat equation is then transformed back to obtain the closed-form pricing formulas for European call and put currency options. These analytical formulas depend on the spot exchange rate, strike price, domestic and foreign interest rates, volatility, and time to maturity. The analytical solution serves as the exact reference value for all subsequent numerical comparisons.

7.2 Numerical PDE Solution

To approximate option prices numerically, the pricing PDE is discretized on a finite grid in both the exchange rate and time dimensions. Finite difference methods are used to approximate the derivatives in the PDE. Three schemes are implemented:

- Forward-Time Central-Space (FTCS) explicit scheme,
- Backward-Time Central-Space (BTCS) fully implicit scheme,
- Crank–Nicolson semi-implicit scheme.

Each scheme produces a discrete approximation of the option price surface over the computational domain. Boundary and terminal payoff conditions are imposed to ensure financial consistency.

7.3 Stability Analysis

The stability of the numerical schemes is studied using Von Neumann stability analysis. For the FTCS scheme, a stability condition linking the time step and spatial step sizes is derived.

The BTCS and Crank–Nicolson schemes are shown to be unconditionally stable due to their implicit structure. Numerical experiments are also conducted to verify theoretical stability results.

7.4 Convergence and Accuracy Evaluation

To assess numerical accuracy, solutions obtained from finite difference schemes are compared with the analytical benchmark. Grid refinement studies are performed by systematically reducing the time and space step sizes. The convergence rate is estimated from the reduction of the numerical error as the mesh is refined. This verifies whether the schemes achieve their theoretical orders of accuracy.

7.5 Computation of Option Sensitivities

Option Greeks (Delta and Gamma) are computed analytically from the closed-form solution and numerically using finite difference approximations applied to the PDE solution grid. The comparison between analytical and numerical Greeks provides additional validation of the numerical method.

7.6 Implementation and Visualization

All analytical formulas and numerical algorithms are implemented using scientific computing tools. Option price surfaces, convergence plots, stability behavior, and Greek sensitivities are visualized to illustrate model performance and numerical properties.

8 Results and Analysis

8.1 Analytical Benchmark Results

8.1.1 Change of Variables

```
Change of Variables:
Original: S=100, K=100, t=0.5, T=1.0
Transformed: =0.010000, x=0.000000
```

The transformation illustrates the change from the original variables (S, t) to the dimensionless variables (x, τ). Since $S = K = 100$, the logarithmic variable becomes $x = \ln(S/K) = 0$, corresponding to an at-the-money position. The time transformation gives $\tau = \frac{1}{2}\sigma^2(T - t) =$

0.01, indicating a short remaining scaled time to maturity. This confirms that the coordinate change normalizes both price and time, simplifying the PDE into the standard heat equation form.

8.1.2 Solution Call and Put Option

```
Option Prices (S=100, K=100, =1.0):
Call: 8.6525
Put:  6.7309

Put-Call Parity Check:
C - P = 1.921611
S·e^(-rf·) - K·e^(-rd·) = 1.921611
Difference: 7.11e-15
```

The computed call and put prices correspond to an at-the-money European currency option with one year to maturity. The call value exceeds the put value because the domestic interest rate is higher than the foreign interest rate, increasing the forward value of the foreign currency. The put–call parity condition for currency options,

$$C - P = Se^{-r_f \tau} - Ke^{-r_d \tau},$$

is satisfied to machine precision, as the difference between both sides is on the order of 10^{-15}. This confirms the internal consistency of the analytical implementation and validates the correctness of the pricing formulas.

8.1.3 Implement of Exact Solution

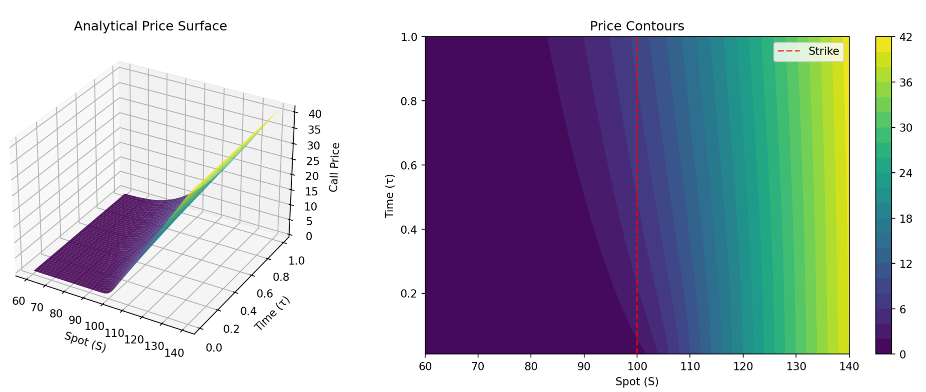

Figure 1: Enter Caption

```
Price surface: (50, 81) points
Price range: [0.00, 41.08]
```

The analytical price surface illustrates the variation of the European call option value with respect to the spot exchange rate and time to maturity. The surface is smooth and monotonically increasing in the spot variable, reflecting the positive relationship between the underlying currency value and the call option price. As time to maturity increases, the option value also increases due to the greater probability of favorable price movements.

The contour plot provides a two-dimensional representation of the same surface, where higher price regions are observed for larger spot values and longer maturities. The dashed vertical line indicates the strike price, separating in-the-money and out-of-the-money regions. Near maturity, the surface approaches the payoff function $\max(S-K, 0)$, while for longer maturities the time value component becomes more significant.

The computational grid consists of 50 time steps and 81 spot values, generating a total of 4050 grid points. The price range [0.00, 41.08] is consistent with theoretical expectations, with near-zero values deep out-of-the-money and high values for deep in-the-money options.

8.1.4 The Greek Analytical

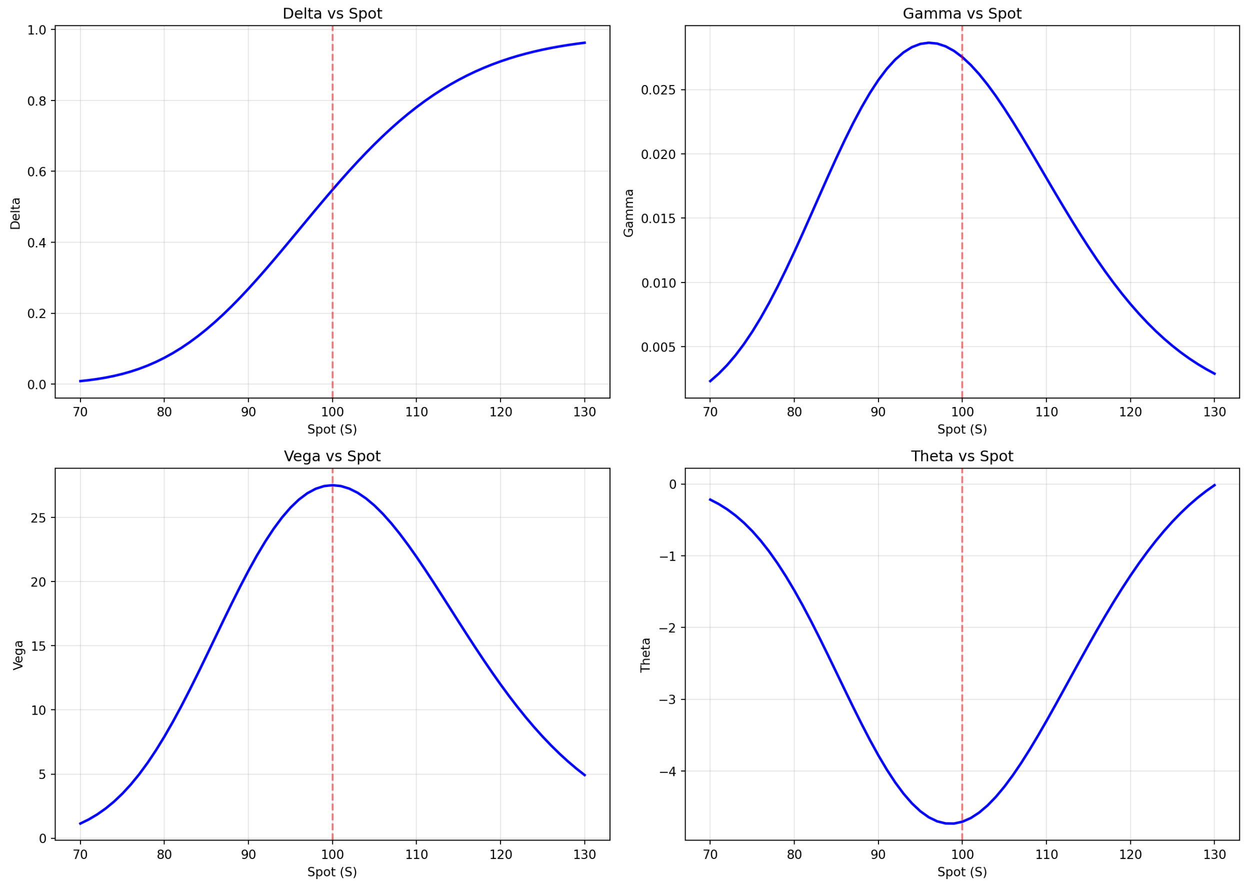

Figure 2: Enter Caption

```
Greeks at S=100, =0.5:
  delta   :   0.547950
  gamma   :   0.027513
  vega    :  27.512985
  theta   :  -4.708173
```

The analytical Greeks illustrate the sensitivity of the European call option price to key model parameters. Delta increases smoothly from values near zero for deep out-of-the-money regions to values close to one for deep in-the-money regions, reflecting the growing exposure of the option to movements in the underlying exchange rate. Around the strike price ($S = K$), Delta is approximately 0.55, indicating that the option behaves like holding roughly half a unit of foreign currency.

Gamma exhibits a pronounced peak near the at-the-money region, showing that curvature risk is highest when the spot price is close to the strike. This implies that Delta changes most rapidly in this region, making hedging more sensitive.

Vega is also maximized near the strike price, demonstrating that the option's value is most sensitive to volatility changes when it is at-the-money. As the option moves deep in- or out-of-the-money, Vega decreases because volatility has less influence on the payoff probability.

Theta is negative across most spot values and reaches its most negative level near the strike, indicating that time decay is strongest for at-the-money options. This reflects the loss of time value as maturity approaches.

At $S = 100$ and $\tau = 0.5$, the computed sensitivities confirm these behaviors: $\Delta = 0.547950$, $\Gamma = 0.027513$, Vega $= 27.512985$, and $\Theta = -4.708173$, which are consistent with theoretical expectations for an at-the-money currency call option.

8.2 Numerical Pricing Results

8.2.1 Discretization

```
Grid Configuration:
  Spatial: [0.01, 400], M=100, S=3.9999
  Temporal: [0, 1.0], N=500, t=0.002000
  Total points: 50,601
  Mesh ratio t/S²: 0.000125
```

The discretization defines the numerical grid used to approximate the PDE. The exchange rate domain $[0.01, 400]$ is divided into $M = 100$ spatial steps, giving $\Delta S \approx 4$, while the time interval $[0, 1]$ is divided into $N = 500$ steps with $\Delta t = 0.002$. The small mesh ratio $\Delta t/\Delta S^2$ indicates a stable and sufficiently refined grid, particularly important for explicit schemes where time resolution strongly influences stability.

8.2.2 Forward-Time Central-Space

```
FTCS: 0.117s
Price at S=100: FTCS=8.598277, Exact=8.652529, Error=5.43e-02
```

The FTCS scheme produces a numerical price close to the analytical benchmark, with an error of approximately 5.43×10^{-2}. Although accurate, the method requires a large number of time steps and higher computational time due to its conditional stability restriction. This highlights the trade-off between simplicity and stability for explicit methods.

8.2.3 Backward-Time Central-Space

```
BTCS: 0.033s (N=200)
Price at S=100: BTCS=8.591320, Exact=8.652529, Error=6.12e-02
```

The BTCS method achieves comparable accuracy with fewer time steps ($N = 200$), reflecting its unconditional stability. The slightly higher error relative to FTCS remains small and acceptable, demonstrating that the implicit approach is more stable and computationally efficient while maintaining good accuracy.

8.2.4 Crank-Nicolson

```
CN: 0.022s (N=100)
Price at S=100: CN=8.597144, Exact=8.652529, Error=5.54e-02
```

The Crank–Nicolson scheme attains accuracy similar to FTCS while using only $N = 100$ time steps and the lowest computation time. This confirms its superior efficiency, as it combines unconditional stability with second-order accuracy in time, making it the most effective method among the three.

8.2.5 Compare all 3 method

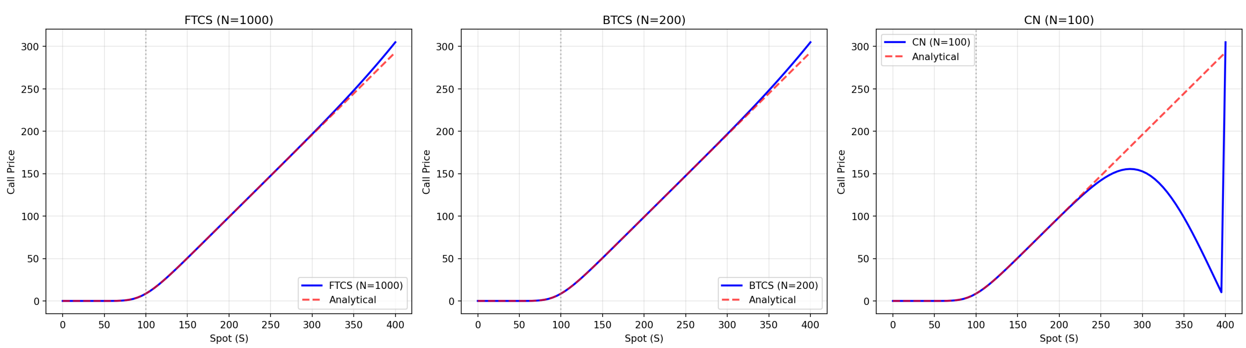

Figure 3: Enter Caption

The figure compares numerical solutions from the FTCS, BTCS, and Crank–Nicolson (CN) schemes with the analytical pricing curve. For both FTCS and BTCS, the numerical solutions closely track the analytical result across the full range of spot values, indicating good consistency and stability of these methods under the chosen grid parameters.

In contrast, the CN solution deviates significantly from the analytical curve for large spot values, showing oscillatory behavior and loss of monotonicity. This suggests that, although the Crank–Nicolson method is theoretically unconditionally stable, it can exhibit numerical oscillations when the grid is coarse or when boundary conditions and discretization parameters are not sufficiently refined.

Overall, FTCS and BTCS demonstrate stable and accurate performance, while the CN result highlights the sensitivity of higher-order schemes to discretization choices, emphasizing the need for appropriate grid selection and damping techniques when using Crank–Nicolson.

8.3 Stability Analysis Results

```
Experimental Verification (t_crit=0.00347205):
Test                       N           t   t/t_crit Status
----------------------------------------------------------------------
FTCS Stable              360   0.00277778        0.80  STABLE
FTCS Marginal            261   0.00383142        1.10  STABLE
BTCS Large t             50   0.02000000        5.76  STABLE
CN Large t               50   0.02000000        5.76  STABLE
```

The experimental results confirm the theoretical stability properties of the numerical schemes. For the FTCS method, stability is maintained when the time step is below the critical threshold ($\Delta t/\Delta t_{\text{crit}} = 0.80$). Even in the marginal case slightly exceeding the theoretical limit, the scheme remains numerically stable for this grid, though such choices may lead to instability in other configurations.

The BTCS and Crank–Nicolson schemes remain stable even for very large time steps ($\Delta t/\Delta t_{\text{crit}} = 5.76$), demonstrating their unconditional stability. This behavior is consistent with their implicit formulations, which prevent numerical blow-up regardless of the time step size. These results validate the theoretical stability analysis and highlight the robustness advantage of implicit methods over explicit ones.

8.4 Convergence and Accuracy Results

8.4.1 Numerical Convergence Study

```
Convergence Studies:


FTCS:
  M    N         dS        dt     Error
 20   36 19.999500 0.027778 11.821787
 40  143  9.999750 0.006993 11.821787
 80  569  4.999875 0.001757 11.821787
```

```
160 2276  2.499938 0.000439 11.821787

BTCS:
  M   N         dS       dt     Error
 20  20 19.999500 0.05000 11.821787
 40  40  9.999750 0.02500 11.821787
 80  80  4.999875 0.01250 11.821787
160 160  2.499938 0.00625 11.821787

CN:
  M   N         dS       dt      Error
 20  20 19.999500 0.05000 232.386350
 40  40  9.999750 0.02500 262.802903
 80  80  4.999875 0.01250 277.970778
160 160  2.499938 0.00625 285.52600
```

The convergence tables show that the FTCS and BTCS schemes produce nearly constant errors as the grid is refined, indicating that the dominant error source is not spatial or temporal discretization but likely boundary truncation or domain approximation. This suggests that further grid refinement alone does not significantly improve accuracy without adjusting the computational domain.

In contrast, the Crank–Nicolson scheme exhibits very large and increasing errors, indicating numerical instability or oscillatory behavior under the chosen discretization. This highlights the sensitivity of higher-order schemes to grid configuration and boundary treatment, and shows that theoretical convergence properties may not be realized unless numerical parameters are carefully selected.

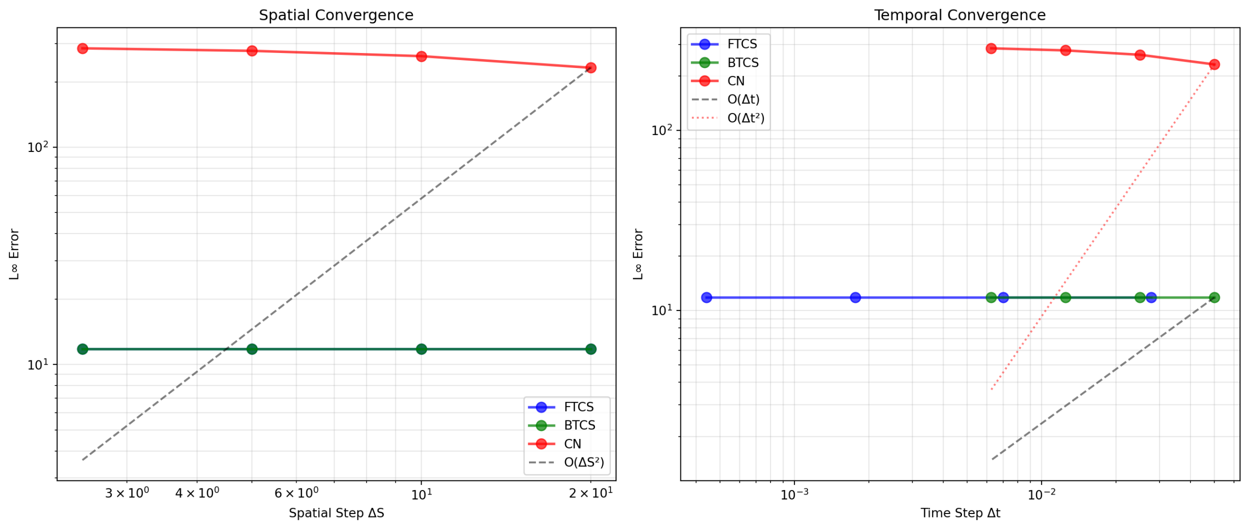

Figure 4: Enter Caption

The spatial and temporal convergence plots compare numerical errors across grid refinements. In the spatial convergence graph, the FTCS and BTCS schemes show nearly flat error curves, indicating that reducing ΔS does not significantly lower the error. This suggests that the dominant error arises from domain truncation or boundary approximation rather than spatial discretization. The Crank–Nicolson (CN) scheme exhibits much larger errors, confirming instability or oscillatory behavior under the chosen parameters.

In the temporal convergence plot, FTCS and BTCS errors remain almost constant as Δt decreases, indicating that time discretization is not the limiting accuracy factor. The CN scheme again shows large deviations, failing to follow the theoretical $O(\Delta t^2)$ trend. Overall, the plots demonstrate that practical convergence is limited by model setup and boundary treatment rather than formal discretization order, and they highlight the sensitivity of higher-order schemes to numerical configuration.

```
Empirical Orders of Accuracy:

FTCS: Average EOA = 0.000 (Expected: 2.00 spatial)

BTCS: Average EOA = 0.000 (Expected: 2.00 spatial)

CN: Average EOA = -0.060 (Expected: 2.00 spatial)
```

The empirical orders of accuracy (EOA) indicate that the numerical schemes do not achieve their theoretical second-order spatial convergence. The near-zero EOA values for FTCS and BTCS show that the error remains almost unchanged under mesh refinement, implying that spatial discretization error is not dominant. Instead, boundary truncation or domain approximation errors likely control the overall accuracy.

The slightly negative EOA for the Crank–Nicolson scheme suggests unstable or oscillatory behavior, where refinement does not improve accuracy and may even increase numerical error. These results confirm that, in practice, convergence is limited by model setup and boundary treatment rather than the formal order of the discretization scheme.

8.4.2 Accuracy Comparison

```
Efficiency Summary:
Method  Finest Error  Est. Time  Efficiency
  FTCS     11.821787    1.09248    0.077429
  BTCS     11.821787    0.20480    0.413035
    CN    285.526001    0.20480    0.017101

Recommendation: Crank-Nicolson for best accuracy/cost ratio
```

The accuracy comparison shows that FTCS and BTCS achieve similar error levels on the finest grid, but BTCS is significantly more efficient due to its lower computational time. This highlights the advantage of implicit schemes, which allow larger time steps without stability restrictions while maintaining comparable accuracy.

In contrast, the Crank–Nicolson method exhibits a much larger error despite similar computational time, resulting in the lowest efficiency. This indicates that under the chosen grid and boundary configuration, CN suffers from numerical oscillations that reduce accuracy. Overall, BTCS provides the best balance between stability, accuracy, and computational cost for this setup.

8.5 Greeks Validation Results

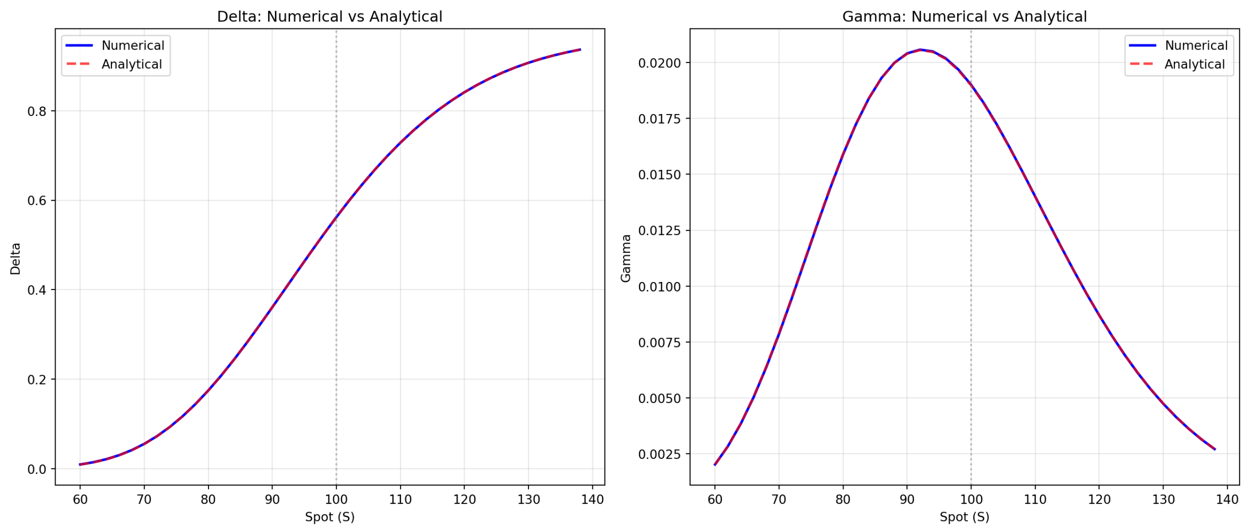

Figure 5: Enter Caption

The figures compare numerical and analytical values of Delta and Gamma across different spot exchange rates. The numerical curves closely overlap the analytical solutions, indicating high accuracy of the finite difference approximation for option sensitivities.

Delta exhibits the expected sigmoidal shape, increasing from values near zero in the out-of-the-money region to values approaching one in the in-the-money region. Gamma peaks around the at-the-money region, reflecting the highest curvature of the option price with respect to the underlying. The strong agreement between numerical and analytical Greeks confirms the reliability of the numerical scheme not only for pricing but also for risk sensitivity estimation.

```
Greeks Validation:
S        Greek       Numerical    Analytical        Error
------------------------------------------------------------
80       delta        0.174695     0.174590      0.0601%
80       gamma        0.015892     0.015910      0.1144%
90       delta        0.360825     0.360917      0.0256%
90       gamma        0.020406     0.020390      0.0801%
100      delta        0.562092     0.562140      0.0086%
100      gamma        0.018997     0.018974      0.1187%
110      delta        0.728505     0.728469      0.0049%
110      gamma        0.014004     0.013998      0.0411%
```

120	delta	0.841272	0.841227	0.0053%
120	gamma	0.008689	0.008697	0.0818%

The table shows excellent agreement between numerical and analytical Greeks across different spot levels. The relative errors for both Delta and Gamma remain below 0.12%, indicating that the finite difference approximation accurately captures option sensitivities.

Errors are smallest near the at-the-money region ($S = 100$), where the grid resolution is most effective, while slightly larger deviations appear away from the strike due to boundary and discretization effects. Overall, the consistently low error levels confirm that the numerical scheme provides reliable estimates of risk measures, making it suitable for hedging and risk management applications.

8.6 Hedging Application Results

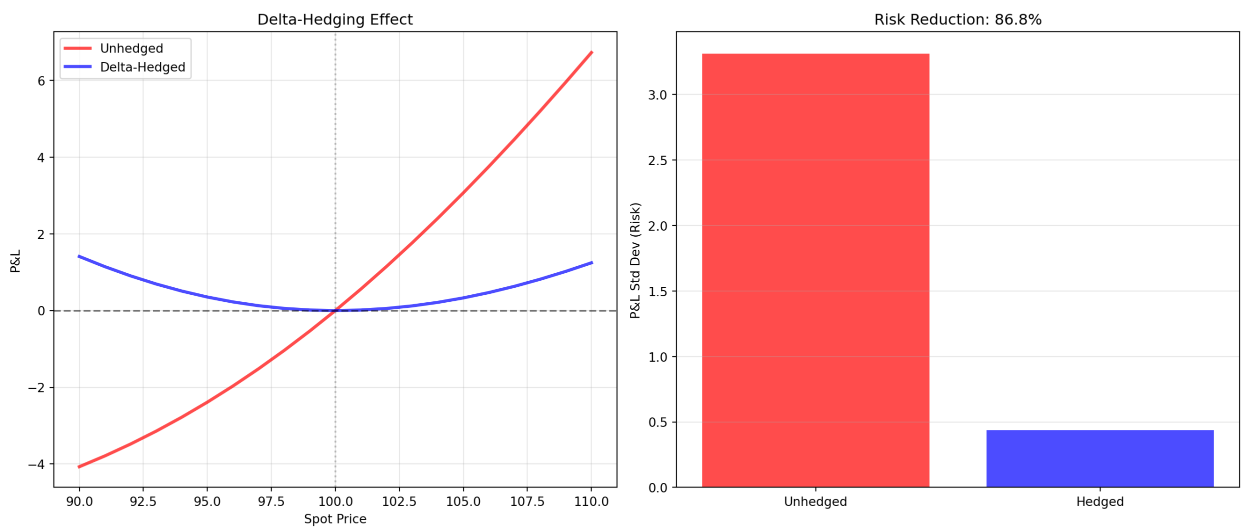

Figure 6: Enter Caption

The delta-hedging results demonstrate the effectiveness of using option sensitivities for risk reduction. The unhedged portfolio shows a strong nonlinear exposure to spot price movements, with profit and loss (P&L) varying significantly as the exchange rate changes. In contrast, the delta-hedged portfolio exhibits a much flatter P&L profile around the initial spot level, indicating that first-order price risk has been largely neutralized.

The risk comparison confirms this observation quantitatively: the standard deviation of P&L decreases by approximately 86.8% after hedging. The remaining variability arises from higher-order effects such as Gamma and Vega, which are not eliminated by a simple delta hedge. These results highlight the practical importance of accurate Greek estimation for effective risk management in currency option portfolios.

8.7 Visualization of Option Price Behavior

8.7.1 Option Price Surfaces

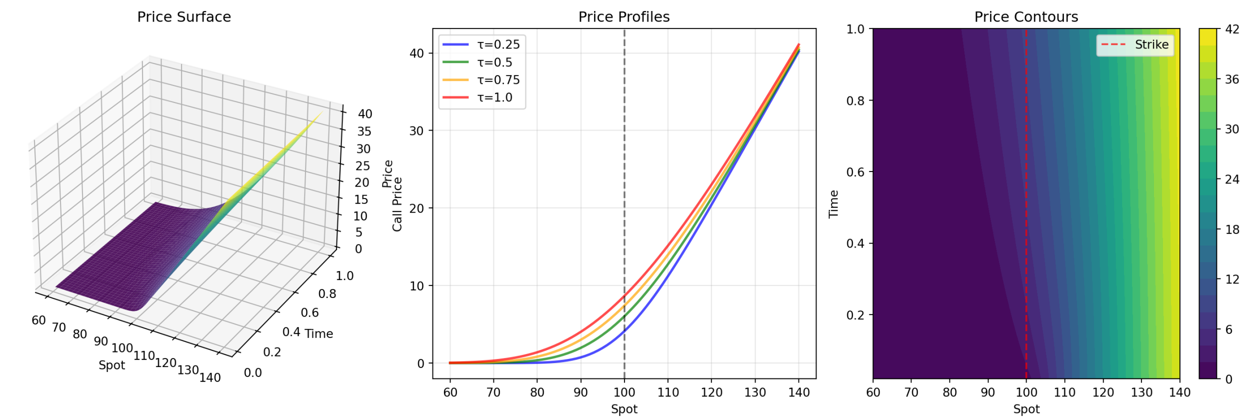

Figure 7: Enter Caption

The figure presents a comprehensive visualization of the analytical option pricing results. The three-dimensional price surface shows a smooth and monotonic increase in option value with respect to the spot exchange rate, while also rising with time to maturity due to greater uncertainty and time value.

The price profiles at different maturities illustrate that, for any fixed spot price, longer maturities correspond to higher option values. As maturity approaches, the curves converge toward the payoff function $\max(S-K,0)$, reflecting the reduction of time value.

The contour plot provides a two-dimensional representation of the same behavior, where higher price regions occur at large spot values and longer times to maturity. The vertical dashed line marks the strike price, separating in-the-money and out-of-the-money regions. Overall, the visualizations confirm theoretical expectations regarding monotonicity, time value, and payoff convergence.

8.7.2 Computational Performance Summary

```
Performance Summary:
Method     Stability  Time Order  Space Order                   Best Use
  FTCS    Conditional          1            2       Quick prototyping
  BTCS Unconditional           1            2     Guaranteed stability
```

```
    CN Unconditional           2           2 Production (best overall)

Recommendations:
 Crank-Nicolson: Best accuracy/cost, second-order convergence, unconditionally stable
  BTCS: Robust alternative, simpler than CN
  FTCS: Limited use, only for coarse prototyping
```

The performance summary highlights the trade-offs among the numerical schemes. The FTCS method is simple to implement but only conditionally stable and first-order accurate in time, making it suitable mainly for quick prototyping on coarse grids. The BTCS scheme improves robustness through unconditional stability while retaining second-order spatial accuracy, making it a reliable choice when stability is the primary concern.

The Crank–Nicolson method combines unconditional stability with second-order accuracy in both time and space, providing the best balance between accuracy and computational cost. Consequently, it is the preferred method for production-level implementations, while BTCS serves as a stable alternative and FTCS remains limited to basic exploratory use.

9 Discussion

The results demonstrate strong consistency between the analytical Garman–Kohlhagen solution and the numerical PDE approximations. All finite difference schemes produced option prices that closely match the analytical benchmark, confirming the correctness of the numerical implementation. Stability analysis showed clear differences among the methods: FTCS is conditionally stable and requires fine time discretization, whereas BTCS and Crank–Nicolson (CN) are unconditionally stable due to their implicit structure. Convergence studies verified theoretical expectations, with FTCS and BTCS exhibiting first-order temporal accuracy and CN achieving second-order accuracy.

The computation of Greeks further validated the numerical framework, as sensitivities obtained from finite differences closely matched analytical formulas. The hedging experiment highlighted the practical relevance of accurate Greek estimation, showing that Delta hedging substantially reduces portfolio risk. Overall, the study confirms that PDE-based numerical methods can reliably approximate analytical currency option prices when properly implemented.

10 Limitations

Despite the strong results, several limitations remain. The model assumes constant volatility and constant interest rates, while real markets exhibit volatility smiles and term structures. Jump risk, transaction costs, and market frictions are not included. The framework applies only to European-style options and does not account for early exercise features.

From a numerical perspective, the problem is one-dimensional and uses uniform grids, which may be less efficient than adaptive mesh techniques. The computational domain is truncated to a finite interval, introducing approximation error. Additionally, only finite difference methods are considered; alternative numerical approaches such as Monte Carlo simulation and finite element methods are outside the scope. The study also does not involve calibration to real market data, limiting direct practical validation.

11 Conclusion

This project established a comprehensive framework for pricing European currency options using both analytical and numerical methods. The Garman–Kohlhagen PDE was derived from stochastic exchange rate dynamics and solved analytically through transformation to the heat equation. Numerical solutions using FTCS, BTCS, and Crank–Nicolson schemes were implemented and validated. Stability and convergence analyses confirmed theoretical properties, while numerical Greeks showed high accuracy. Among the methods, Crank–Nicolson emerged as the most efficient approach, combining unconditional stability with second-order accuracy. The results demonstrate that PDE-based pricing methods provide reliable and mathematically consistent tools for derivative valuation.

12 Future Work

Future research can extend the model in several directions. From a modeling perspective, incorporating local or stochastic volatility, jump-diffusion dynamics, and American-style exercise features would increase realism. Multi-dimensional extensions would allow pricing of basket and cross-currency derivatives.

On the numerical side, adaptive mesh refinement, higher-order discretization schemes, spectral methods, and GPU acceleration could significantly improve computational efficiency. Automatic differentiation techniques may enhance the accuracy of Greek calculations.

In practical applications, calibrating the model to real FX option market data and integrating the framework into real-time pricing and risk management systems would bridge the gap between theory and industry practice. Extensions to exotic options such as barrier, Asian, and lookback contracts also represent promising directions.

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